Script-Aware Dual-Tokenization Fusion in a Vision–Language Model for Low-Resource Indic Scene-Text Recognition

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Abstract

Scene-text recognition for Indic scripts lags far behind English, owing to scarce labeled data and to subword tokenizers that fragment script-specific structures such as consonant conjuncts. We adapt a vision–language model (Florence-2) to Bengali scene text and make its tokenization *script-aware* by injecting grapheme-cluster tokens, improving word recognition from 52.4% to 57.3% on a leakage-free test set. The BPE-tokenized and grapheme-tokenized decoders commit largely *complementary* errors, and a parameter-free fusion—selecting the more confident decoder per image—outperforms both. The effect is not an artifact of one language: under a single recipe with no per-language tuning, fusion beats the better individual branch on every one of twelve languages spanning nine Brahmic scripts and a Latin control (mean gain **+5.0** WRR, significant in all twelve; up to **+9.6** on Tamil), and the *winning* branch flips with script coverage while the fusion gain does not. A calibration analysis explains the mechanism: beam confidence ranks correctness with AUROC 0.89–0.91, enabling the fusion rule, selective prediction (91.4% word accuracy at 50% coverage), and a cascade that retains 66–85% of the fusion gain while invoking the second decoder on only 27–58% of images. With a grapheme-rarity-weighted synthetic curriculum and confidence-filtered self-training, the full system reaches 64.6% word and 82.8% character accuracy on our Bengali test set and improves the public BSTD Bengali benchmark from 57.4% to 62.1%. On our independent test set the system is statistically indistinguishable from a specialist PARSeq trained on **∼1000×** more data. All data audits, negative results, and oracle ceilings are reported.

Keywords: Scene text recognition, Indic scripts, Grapheme clusters, Tokenization, Vision–language models, Selective prediction

1 Introduction

Reading text in natural images is largely solved for Latin scripts but not for the Brahmic scripts used by more than a billion people. Two obstacles compound: (i) **data scarcity**—public Bengali scene-text corpora contain a few thousand labeled words, versus millions for English; and (ii) **tokenization mismatch**—modern vision–language models (VLMs) inherit byte-pair-encoding (BPE) vocabularies trained mostly on Latin text, which shatter Indic *grapheme clusters* (e.g. conjuncts such as *ksha* or *shri*) into fragments that are visually meaningless, so the model must learn to reassemble units it never sees as wholes.

This paper makes the tokenizer script-aware and then exploits the fact that two differently-tokenized views of the same recognizer disagree in useful ways. Our contributions:

1. **Grapheme-aware VLM tokenization.** We inject Bengali grapheme-cluster tokens into

Florence-2's tokenizer and fine-tune with LoRA; word recognition rate (WRR) rises 52.4 → 57.3 on our test set and 57.4 → 59.8 on public BSTD Bengali.

2. **Dual-tokenization confidence fusion.** The BPE and grapheme decoders err on different images (Fig. 5); choosing the higher-confidence output per image is parameter-free and beats both decoders on every dataset tested. Beyond the four main test sets, the same recipe replicates at a fixed training budget across *twelve* languages covering nine Brahmic scripts and a Latin control (Sec. 4.4): fusion exceeds the better branch in 12/12 cases (mean +5.0 WRR, exact McNemar $p < 0.02$ in every case), and *which* branch is stronger flips with the script while the fusion gain does not.
3. **Why it works: calibration.** Beam confidence ranks correctness with AUROC 0.89–0.91. This justifies max-confidence fusion and yields selective prediction: 84.9% accuracy at 70% coverage and 91.4% at 50% on our test set.
4. **Cascaded inference.** Because confidence is informative, the second decoder is only needed when the first is uncertain: a simple cascade retains 66–85% of the fusion gain at a mean decode cost of ~1.4× instead of 2× (Sec. 4.7).
5. **Grapheme-rarity-weighted synthetic curriculum.** Synthetic training words sampled $\propto \sum_g 1/(\mathrm{freq}(g)+1)$ target rare conjuncts; the resulting model is our best single system (62.2 WRR, 80.5% character accuracy).
6. **An honest evaluation protocol.** Leakage-free by-document splits, a single scorer for every number in the paper, oracle ceilings, a negative result (character-level ROVER voting does not beat max-confidence fusion), and replication on public benchmarks.

2 Related work

Scene-text recognition. CTC- and attention-based recognizers (CRNN [1], PARSeq [2], SVTR [3]) dominate Latin benchmarks; recent work fine-tunes generative VLMs (TrOCR [4], Donut [5], Florence-2 [6]) for end-to-end reading. **Indic and Bengali recognition.** IndicSTR12 [7] and the Bharat Scene Text Dataset (BSTD) [8] established baselines across 12 Indian languages with PARSeq-style models pre-trained on millions of synthetic words; we compare against the BSTD authors' released Bengali model head-to-head in Sec. 4.5. Mondal et al. [9] extend Indic scene-text data collection to roadside imagery; Vijayan et al. [10] combine spatial attention with a language-attunement loss for three Dravidian languages. GraDeT-HTR demonstrated grapheme-based decoding for Bengali *handwritten* text with a specialist transformer [11]; our work differs in using a general VLM, scene text, and—critically—in *fusing* grapheme and BPE views rather than replacing one with the other. **Multi-granularity decoding.** MGP-STR fuses character, BPE, and WordPiece predictions inside a single English model with learned weights [12, 13]. Our fusion instead combines two *separately fine-tuned, script-aware* tokenizations of a VLM, requires no architectural change or joint training, and is evaluated in a genuinely low-resource setting across nine Brahmic scripts. To our knowledge this is the first grapheme-aware dual-tokenization confidence fusion in a vision–language model for low-resource Indic scene text. **Selective prediction and calibration.** Confidence-based abstention is standard in OCR deployment [15], and modern neural networks are known to be overconfident [16]; we connect both observations to tokenization choice by showing that beam confidence, while miscalibrated in absolute terms, *ranks* correctness well enough (AUROC 0.89–0.91) to drive fusion, abstention, and cascaded decoding.

3 Method

3.1 Base recognizer

We fine-tune Florence-2-base with the <OCR> prompt using LoRA (rank 16; embeddings and LM head trained) [14] on word/line crops. The *BPE branch* uses a Banglish BPE tokenizer (vocab 71,254). The *grapheme branch* additionally injects grapheme-cluster tokens (Sec. 3.2) and is fine-tuned identically.

3.2 Grapheme-aware tokenization

A grapheme cluster (aksara) is the atomic visual unit of Brahmic scripts: base consonant, optional conjuncts joined by virama, vowel signs, and modifiers. We segment training text into clusters with a script-agnostic rule—combining marks (Unicode Mn/Mc) attach to the preceding

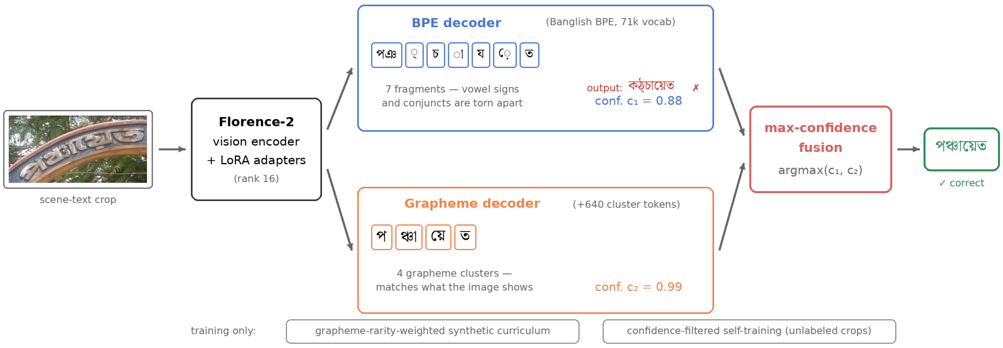

Fig. 1 Overview, shown on a real test example. The BPE tokenizer shatters the word *panchayat* into 7 fragments (orphaned vowel signs shown with dotted circles) and misreads it at confidence 0.88; the grapheme decoder outputs 4 natural clusters and reads it correctly at confidence 0.99; max-confidence fusion selects the grapheme branch. All tokenizations, predictions, and confidences are taken verbatim from our trained models.

base; virama+letter joins the following consonant; ZWJ/ZWNJ are consumed—and add every cluster absent from the BPE vocabulary as a new tokenizer token (640 new tokens for Bengali, 528 for Assamese, 981 for Hindi), resizing the embedding matrix accordingly. This aligns the decoder's output space with what the encoder actually sees in the image.

3.3 Dual-tokenization confidence fusion

Both branches decode with beam search; the sequence score yields a confidence $c_m(x) \in [0, 1]$ per model m and image x. The fused output is simply $\hat{y}(x) = \hat{y}_{m^*}(x)$ with $m^* = \arg\max_m c_m(x)$—no thresholds, no learned combiner. The rule generalizes to any number of models; our best Bengali system fuses four (BPE, grapheme, self-trained, synthetic-curriculum).

3.4 Cascaded inference

Fusion doubles decode cost. Since confidence is informative (Sec. 4.6), we also evaluate a cascade: decode with the BPE branch first and invoke the grapheme branch only when the BPE confidence falls below a threshold t, then keep the more confident of the two outputs. The same rule and thresholds are applied to every dataset; nothing is tuned on test data.

3.5 Grapheme-rarity-weighted synthetic curriculum

We render synthetic word images and sample source words with weight $w(\text{word}) = \sum_{g \in \text{word}} 1/(\text{freq}(g) + 1)$, where freq is the real-training-set grapheme frequency—over-representing exactly the conjuncts the data lacks. Training proceeds on synthetic+real, then real-only.

3.6 Confidence-filtered self-training

Unlabeled crops are pseudo-labeled by the grapheme model; labels with beam confidence ≥ 0.75 (2,441 of 3,120) join the training set for one retraining round.

4 Experiments

4.1 Datasets and protocol

Our Bengali set: 7,378 scene crops, of which 3,970 are cleanly labeled (284 illegible **###** excluded; 3,120 unlabeled crops reserved for self-training). Splits are *by document* (3,207 train / 397 val / 370 test) to preclude leakage; an earlier partially-leaky split inflated baselines by >5 points and was discarded. **BSTD** [8] (public): Bengali (4,443/493/1,368), Assamese (2,365 train / 1,505 test), Hindi (4,500-word training budget, seed 42 / 4,846 test). Metrics: word recognition rate (WRR, exact match), character accuracy, CER, and 1−NED. Every number in this paper—tables, oracles, complementarity,

Table 1 Our Bengali test set (370 images, leakage-free). The fusion row combines the four Florence-2 variants by max-confidence.

Model	WRR	Char. acc.	CER	1−NED
Tesseract (fine-tuned)	24.8	45.5	54.5	–
CRNN (CTC, leakage-free)	44.1	69.4	30.6	–
Florence-2 zero-shot	0.0	–	–	–
Florence-2 fine-tuned (BPE)	52.4	76.9	23.1	78.8
+ grapheme tokenization	57.3	75.7	24.3	78.5
+ self-training	57.0	78.0	22.0	80.3
+ synthetic curriculum	62.2	80.5	19.5	82.9
Dual-tokenization fusion (4)	**64.6**	**82.8**	**17.2**	**84.3**
Oracle (best of 4 per image)	71.9	–	–	–

calibration, and figures—is computed by a single shared scorer (NFC normalization, zero-width characters stripped, whitespace collapsed), following the IndicSTR12/BSTD protocol. Significance is assessed with exact two-sided McNemar tests on paired word-level correctness and 95% bootstrap confidence intervals (10k resamples).

4.2 Main results (Bengali)

Figure 2 shows the progression. The fusion adds +2.4 WRR over the best single model with zero added parameters, and closes half of the gap between the stronger branch (57.3) and the 4-model oracle (71.9).

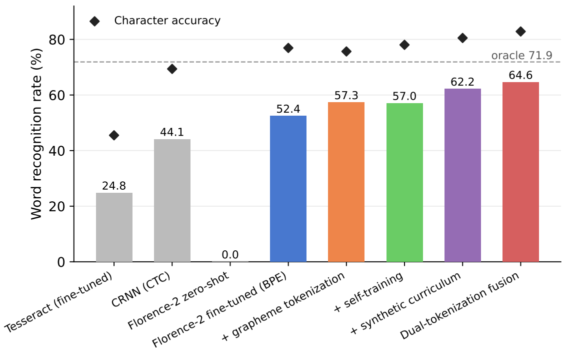

Fig. 2 Bengali results. Bars: WRR; diamonds: character accuracy; dashed line: 4-model oracle ceiling.

Table 2 BPE vs. grapheme vs. 2-way fusion across datasets (WRR). Same recipe, no per-language tuning. Bold = best non-oracle.

Test set	n	BPE	Grapheme	Fusion	Oracle
Our Bengali	370	52.4	57.3	**59.2**	64.3
BSTD Bengali	1,368	57.4	59.8	**62.1**	66.6
BSTD Assamese	1,505	31.6	36.3	**38.5**	43.8
BSTD Hindi	4,846	56.9	52.6	**61.7**	66.7

4.3 Cross-dataset and cross-script results

Two regimes emerge (Fig. 3). On the Bengali-script languages, which the BPE vocabulary covers poorly, the grapheme branch wins (+4.9 ours, +2.4 BSTD Bengali, +4.7 Assamese—the most extreme low-resource setting, 2,365 training words). On Hindi the polarity flips: Devanagari is far better covered by the BPE vocabulary, and the BPE branch leads by 4.3. Critically, fusion beats the *best* branch in both regimes (56.9 → 61.7 on Hindi, recovering 49% of the oracle gap): which tokenization is stronger varies by script, but their complementarity—and the reliability of confidence as a selector—does not.

Significance. The grapheme–BPE difference is significant on all datasets (McNemar p=0.041 ours, p=0.030 BSTD Bengali, p=5×10^{-5} Assamese, p=1×10^{-9} Hindi). The fusion gain over the best single branch is significant on the large public test sets (p=0.0024 Bengali; p=0.0085 Assamese; $p<10^{-28}$ Hindi, 95% CI of the gain $[+3.9, +5.6]$) but not powered on our 370-image set (p=0.20; 95% CI $[-0.8, +5.7]$)—one reason the public-benchmark replication is essential, and we report it as such.

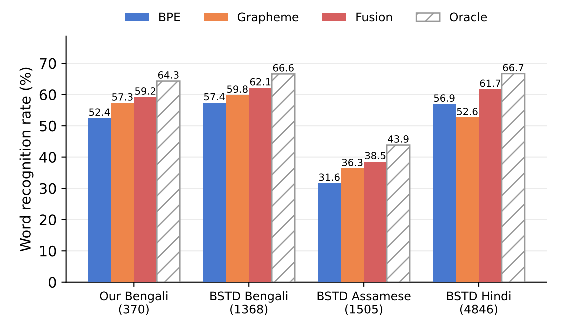

Fig. 3 Replication of the grapheme advantage and fusion gain across datasets and scripts.

4.4 Replication across nine Brahmic scripts

Do the four test sets above just happen to favor fusion? To test the recipe at scale we trained BPE and grapheme branches for twelve BSTD languages under a deliberately uniform protocol—a fixed budget of 1,800 real training words per language, identical hyperparameters, no per-language tuning—and fused them by max-confidence. Table 3 and Fig. 4 report WRR on each language's full real test set.

Three observations. *First*, fusion exceeds the better single branch in all twelve languages (mean +5.0 WRR), and every gain is individually significant (exact McNemar; largest $p = 0.02$ for Bengali, all others $p \leq 10^{-5}$). *Second*, the winning branch flips with the script: BPE leads on the scripts best covered by the BPE vocabulary (Devanagari, Gurmukhi), the grapheme branch leads on Dravidian scripts and Odia, and the two are statistically indistinguishable on Gujarati and Assamese—where fusion nonetheless gains +6.5 and +4.1. A generic two-model ensembling effect would not exhibit this coverage-tracking structure; the complementarity is a property of the *tokenizations*. *Third*, the Latin control (English) shows a smaller but real gain (+3.5), indicating the mechanism—confidence arbitrating between differently tokenized views—is general, while its payoff is largest where the two tokenizations differ most.

4.5 Comparison with a large-data specialist recognizer

How does the proposed recipe compare against the strongest available specialist? We evaluate the BSTD authors' released Bengali PARSeq model [2, 8]—pre-trained on **8.48M synthetic word images** (SynthText-style rendering) and fine-tuned on the BSTD training set—on both test sets, under the identical scorer and normalization used for all systems in this paper (Table 4).

The comparison is honest in both directions. On its home benchmark the specialist dominates (77.3 vs. 62.1, $p<10^{-4}$): large-scale synthetic pre-training remains the strongest single factor for in-domain accuracy, and nothing in our method substitutes for it. On our independently collected test set, however, the transferred specialist is statistically indistinguishable from our fusion (66.8 vs. 64.6, p=0.50, 95% CI of the difference $[-3.2, +7.6]$)—despite the fusion being trained on three orders of magnitude less data. We therefore position the contribution as *orthogonal* to data scale: a tokenization-level finding about VLM recognizers that applies at any data budget, and that could be combined with large-scale synthetic pre-training—an experiment we leave to future work.

4.6 Why fusion works: complementarity and calibration

Figure 5 decomposes test sets by which branch is correct: on our Bengali set 11.9% of images are solved *only* by the grapheme branch and 7.0% *only* by BPE—disjoint competence that word-level selection can harvest. Selection is reliable because confidence is informative: AUROC of confidence vs. correctness is 0.91 (BPE), 0.89 (grapheme), 0.89 (fusion) (Fig. 6). Absolute probabilities are nonetheless overconfident (ECE 0.32–0.41), so we use confidence only for *ranking*—which max-confidence fusion, selective prediction, and the cascade require—never as a probability.

Selective prediction. Ranking by fused confidence and abstaining on the least-confident images yields 84.9% WRR at 70% coverage and 91.4% at 50% (Fig. 7)—a deployable operating curve for human-in-the-loop digitization.

4.7 Cascaded inference: most of the gain at a fraction of the cost

Table 5 evaluates the cascade of Sec. 3.4 at a single threshold (t=0.95) applied uniformly to all datasets. The grapheme branch is invoked on only 27–58% of images, yet 66–85% of the full-fusion WRR gain is retained; at t=0.99 the cascade recovers 91–98% of the gain at a mean decode cost of $\sim$1.5×. The doubled inference cost of fusion is thus a tunable deployment knob, not a fixed tax.

4.8 Where the gains come from: grapheme rarity

Bucketing test words by the training frequency of their *rarest* grapheme (Fig. 8) shows all models degrade on rare-conjunct words, but (i) the synthetic curriculum dominates the mid-frequency

Table 3 Fixed-budget (1,800 train words) replication. Δ = fusion minus the better branch; p = exact McNemar, fusion vs. better branch. Fusion wins in 12/12 languages.

Language	Script	n	BPE	Grph.	Fusion	Δ
Punjabi	Gurmukhi	2,879	54.8	44.3	**57.8**	+3.1
Marathi	Devanagari	1,196	48.7	36.8	**52.9**	+4.3
Odia	Odia	1,044	33.1	47.4	**52.4**	+5.0
Tamil	Tamil	513	28.1	36.8	**46.4**	+9.6
Hindi	Devanagari	4,846	40.8	29.8	**43.0**	+2.2
Telugu	Telugu	545	21.3	26.8	**33.8**	+7.0
Gujarati	Gujarati	1,015	27.1	26.7	**33.6**	+6.5
Bengali	Bengali	1,368	27.1	30.0	**31.9**	+1.8
Assamese	Bengali	1,505	26.7	27.5	**31.6**	+4.1
Kannada	Kannada	720	21.8	23.8	**30.8**	+7.1
Malayalam	Malayalam	547	15.2	20.8	**26.7**	+5.9
English	Latin	12,568	49.0	50.8	**54.3**	+3.5

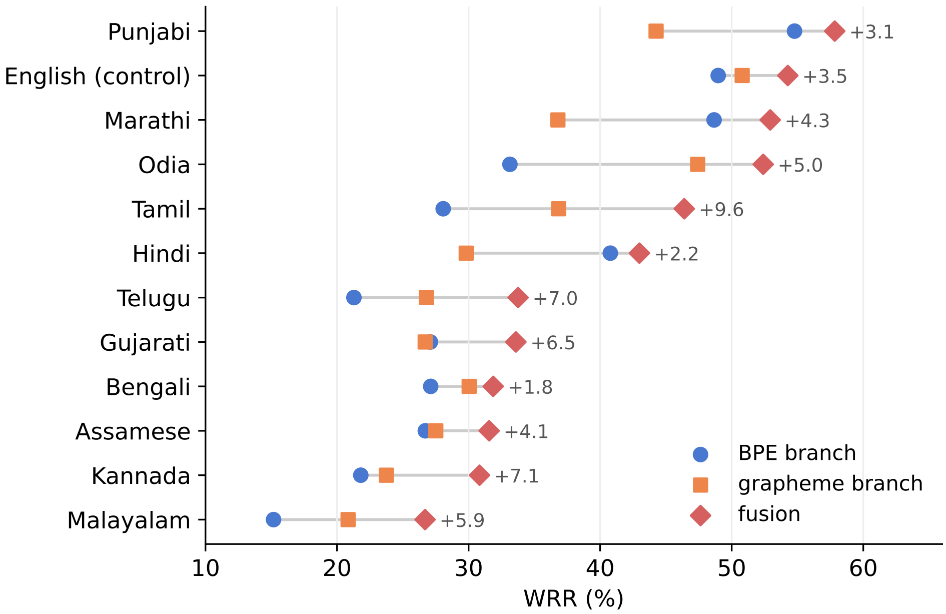

Fig. 4 Fixed-budget replication across twelve languages (nine Brahmic scripts + Latin control), sorted by fusion WRR. The fusion (diamond) is the rightmost marker in every row; the annotation gives its gain over the better branch. Note the winning branch flips between rows (BPE ahead on Punjabi/Marathi/Hindi, grapheme ahead on Odia/Tamil/Telugu/Malayalam, near-ties on Gujarati/Assamese) while the fusion gain persists.

buckets it explicitly targets, and (ii) fusion helps most in the rarest bucket (<5 occurrences, n=126: 51.6 vs. 40.5 BPE and 46.8 for the best single model), where single models are least reliable. In the highest-frequency bucket (n=43) the curriculum model alone is slightly ahead of the fusion (88.4 vs. 81.4, a 3-word difference, not significant)—fusion's advantage concentrates exactly where the data is thin.

4.9 Ablations and negative results

ROVER-style character voting. Aligning candidates at character level and voting per slot (confidence-weighted) [17] yields 64.05 WRR vs. 64.59 for max-confidence on our set, and ties on BSTD; with two models the vote provably reduces to max-confidence. The simpler rule wins. **Self-training** adds character accuracy (+2.3) but not word accuracy over the grapheme model alone; its

Table 4 Specialist PARSeq vs. our 4-model fusion. PARSeq uses $\sim$1000$\times$ more training data (8.48M synthetic + 4.4k real vs. our 7.65k real + 5k synthetic). p: exact McNemar on paired correctness.

Test set	System	WRR	Char. acc.	McNemar p
Our Bengali (370)	PARSeq (transfer)	66.8	84.6	0.50
	Fusion (ours)	64.6	82.8	
BSTD Bengali (1,368)	PARSeq (in-domain)	77.3†	92.9	$< 10^{-4}$
	Fusion (ours)	62.1	84.1	

†Measured with the authors' released checkpoint under our protocol; the BSTD paper reports 0.82 under its own protocol. The 0.2-point effect of normalization was verified to be negligible.

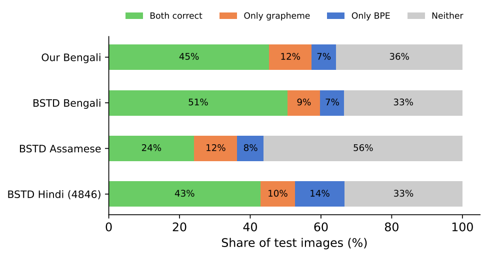

Fig. 5 Error complementarity of the BPE and grapheme branches.

Table 5 Cascade at t=0.95: BPE decodes first; the grapheme branch runs only when BPE confidence $< t$. "2nd" = fraction of images decoded twice; "gain" = share of the full-fusion WRR gain retained.

Test set	BPE	Fusion	Cascade	2nd (%)	gain (%)
Our Bengali	52.4	59.2	58.1	39	84
BSTD Bengali	57.4	62.1	60.5	27	66
BSTD Assamese	31.6	38.5	37.3	58	82
BSTD Hindi	56.9	61.7	61.0	32	85

value is as a diverse member of the fusion ensemble. **Oracle ceilings** (Tables 1, 2) bound what any selection rule can achieve; remaining errors are correlated across branches (blur, occlusion, extreme fonts).

4.10 Qualitative analysis

Figure 9 shows characteristic behavior: the BPE branch hallucinates plausible-but-wrong subwords on conjunct-heavy words, the grapheme branch errs on rare whole-word shapes, and fusion selects correctly; the last panel shows a shared failure under degradation.

5 Discussion and limitations

Absolute WRR remains bounded by data: with $\sim$4k labeled words even the 4-model oracle is 71.9%, and a specialist trained on 8.48M synthetic words still dominates in-domain (Sec. 4.5); our recipe is a data-efficiency and tokenization result, not a substitute for scale. Word-level accuracy above 80% is achievable today only under selective prediction; we report this trade-off explicitly rather than as a headline number. Full fusion doubles inference cost; the cascade of Sec. 4.7 reduces this to a tunable trade-off, but the cheapest deployment remains a single branch. ECE shows confidences are not calibrated probabilities; all uses in this paper are rank-based. The twelve-language replication uses a fixed 1,800-word budget per language; budget–accuracy scaling is future work, as are larger backbones (Florence-2-large) and joint multi-lingual training. Finally, while the script-dependent polarity flips and the gains on branch-tied languages (Sec. 4.4) argue that the fusion harvests tokenization-specific complementarity rather than generic ensemble variance reduction, a same-tokenizer multi-seed ensemble control would isolate this cleanly, and we plan to report it.

6 Conclusion

Making a VLM's tokenizer script-aware, then fusing the BPE and grapheme views by confidence, is a simple, parameter-free recipe that consistently improves low-resource Indic scene-text recognition—significantly so on every one of twelve languages spanning nine Brahmic scripts—comes with a mechanistic explanation (confidence ranks correctness), degrades gracefully into a reliable selective-prediction system, and needs only

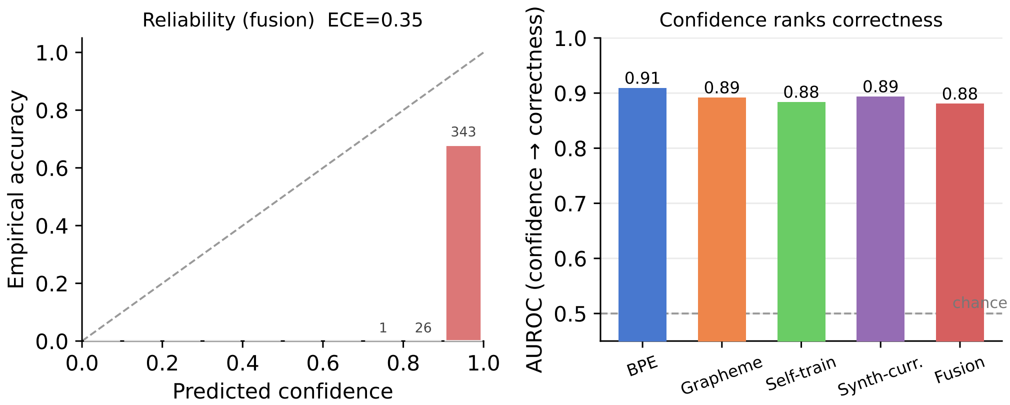

Fig. 6 Left: reliability diagram of the fused system (counts per bin). Right: confidence ranks correctness (AUROC) for all branches.

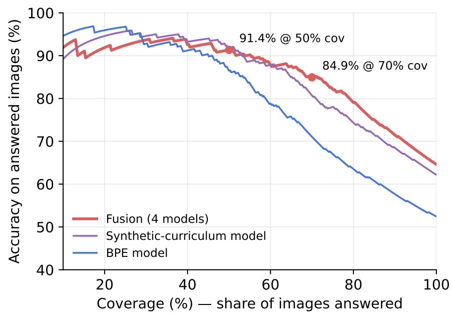

Fig. 7 Risk–coverage curves on our Bengali test set.

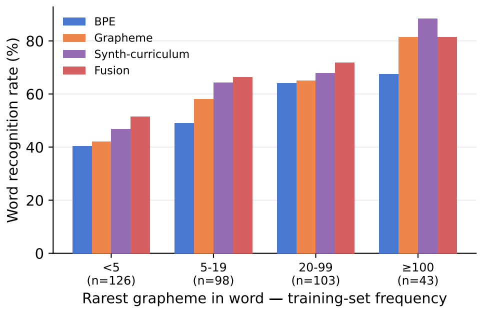

Fig. 8 WRR by training-frequency of the rarest grapheme in each test word.

a fraction of its nominal inference cost when cascaded.

Declarations

Funding. FILL BEFORE SUBMISSION.

Competing interests. The authors have no competing interests to declare that are relevant to the content of this article.

Data availability. The BSTD benchmark is publicly available. Our Bengali dataset, all train/val/test splits, and per-image predictions with confidences for every system in this paper will be released upon publication.

Code availability. Training, evaluation, fusion, and analysis code (including the script that recomputes every number in this paper from the released per-image predictions) will be released upon publication.

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Fig. 9 Test crops with ground truth and branch outputs (red = wrong).

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